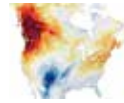




Disease-resistant pigs

Researchers developed pigs resistant to a respiratory illness affecting cattle all over the world. <https://dgit.in/jul21-87>



Pacific NW temperatures rise

Human-driven climate change sent Pacific Northwest temperatures soaring. <https://dgit.in/jul21-88>

DOES FREE Will EXIST?

Can science ever answer that question, and why hasn't it already?

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ho is this Will, and why is he incarcerated? The bad joke at the beginning of the article taken care of,

let's try and understand what free will is before we attempt to decide whether science can prove or disprove its existence. You know a topic is complex when you can't decide on a definition of the topic itself, and free will is the poster child of problematic definitions...

For most people, the definition of free will is the ability of humans to choose between two or more options. This could be the choice between tea and coffee in the morning, or the decision between choosing to kill someone in a rage, or to just walk away. So why does anyone have a problem with this?

Some people define "free will" as total control over one's actions. That one is easy to debunk by merely visiting a doctor who bangs your knee with a rubber mallet and causes your leg to spasm and lift.

Others define free will as the total freedom to do "otherwise". As in, the ability to choose the exact opposite of the choice we made. However, this raises problems. It's like giving people a choice between buttons that light up bulbs in some corner of the globe that are inconsequential to them. They

might choose one button over the other purely randomly. However, tell them that one button lights up a bulb for a poor child studying, while the other lights up a bulb for a chicken incubator to keep eggs warm, and you have introduced complexity that now involves pre-existing biases of the person. All of you already know which button you'd choose, and no, not all of you will agree, because people think differently.

So is free will only exercised when we don't care about the outcome of the choice? We don't know.

While it seems obvious and intuitive to most of us that we all have free will, there are many scientists, philosophers and thinkers who are convinced that we don't. So, why is that?

POLITICS AND RELIGION

Yes, sadly, the two things we usually don't like to talk about at Digit have a lot to do with this topic. We're not going to dwell on it for too long, but the gist of it is that free will brings with it the idea of cause and effect, action and consequences and, most importantly, good and evil. If there is no free will, it means that we are all a victim of our circumstances, and our entire lives, and every decision in those lives, were something that was set in motion at the moment of the Big Bang, and we cannot really be held responsible for our actions. This applies to serial killers and rapists as well as the great humanitarians and heroes of the past, who could not have acted any other way, since they couldn't have "chosen" to change their actions.

As an example, think of a society of murderers. Every member of the society is a murderer, and has to be in order to be accepted. In such a case, someone who does not like the idea of murdering people might lie to be accepted into the society, and merely claim to have murdered someone. In this society, deceit is the bad thing which would be punished, not murder. Weird, isn't it?

NEUROBIOLOGY

When a question like this seeks a scientific answer, it's to those who study human brains for a living that we turn for answers. And, well, they don't seem to have them, even though the camps on either side of the argument are convinced that their side is right.

And let's make a distinction immediately with the functions of the brains. No one ever argues that free will entails the production of blood and repairing of tissues in the body. As far as we know, no matter how hard a conscious human brain wills it, one cannot stop bodily functions just by will alone. There are a myriad unconscious actions happening in our bodies and our brains that are completely automatic and coded into our DNA. Free will, if it exists, is exerted when given a choice of action to our conscious mind. And so, scientists sought to give people those choices and watch how their brains functioned.

Basically the scientific community got involved in this debate based on the data from some studies done by